

ARIMAX Modelling of Australian Residential Property Prices

How do bond yields and unemployment drive the property market and where is it heading?

\$10.5T

Australian Property
Asset Class

192 Qtrs

1978–2025
Dataset

ARIMAX(1,1,2)

Selected
Model

10 Qtrs

Forecast
Horizon

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The Problem & Research Question

Why model property prices with macro drivers?

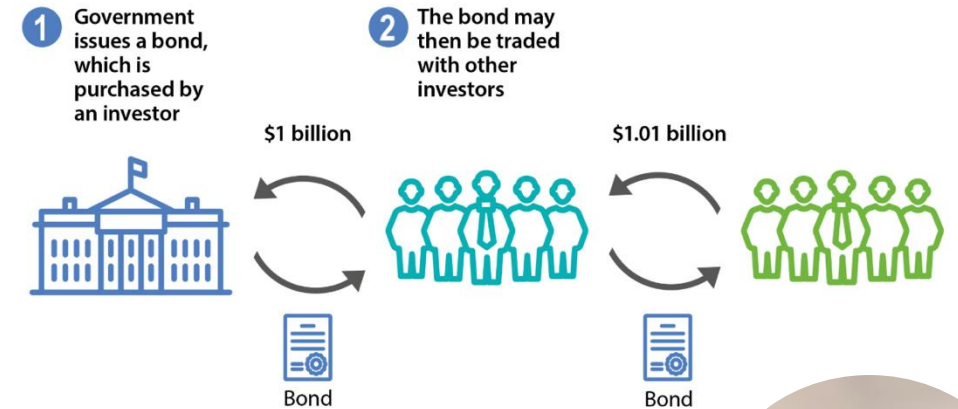
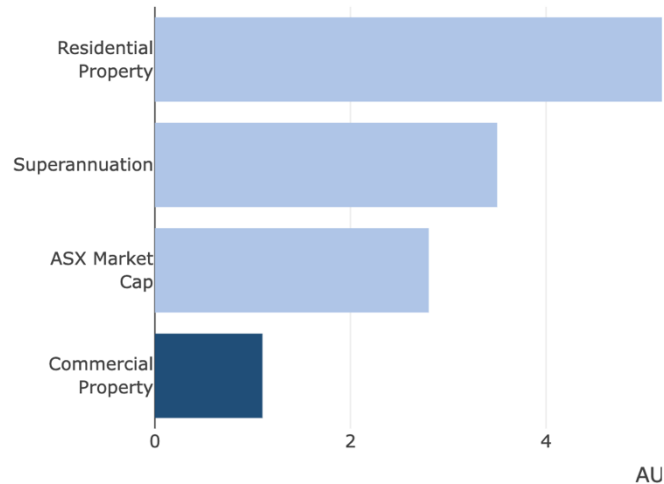
RESEARCH QUESTION

How do bond yields and unemployment help explain movements in Australia's real residential property price index, and what does the fitted ARIMAX model imply for the next 10 quarters?

RPPI

Australian house prices

Inflation adjusted, March 2011 = 100*



Goal: Build a defensible ARIMAX workflow and translate the output into a usable decision tool for investors and property managers.

The Data : Three Series, 47 Years (Q1 1978 - Q4 2025)

192 quarterly observations · No missing values ·
Source: BIS, OECD, ILO via FRED

4× Real Increase

RPPI grew from 35.9 to 141.8 over 47 years long upward trend

Labour Cycles

Unemployment peaked 11.2% (1993) · troughed 3.5% (2022)

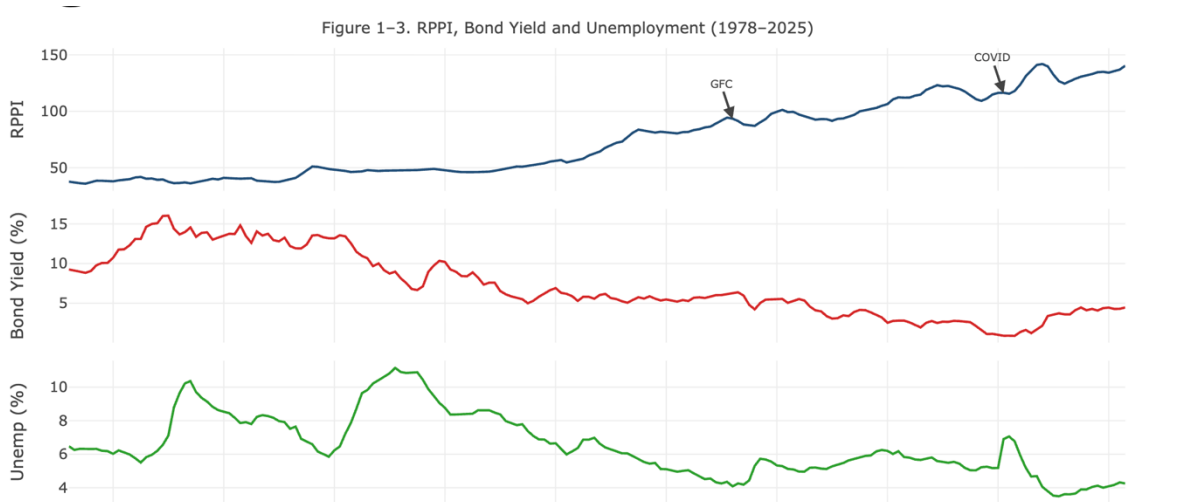
Rate Collapse

Bond yields fell 16% → 0.88% longest decline in AU history

No Missing Data

192 complete quarterly obs after merging all three sources

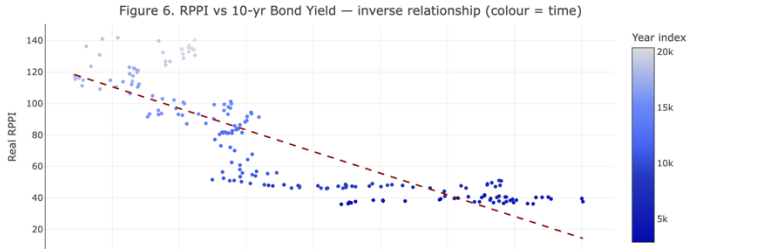
Table 1. Summary statistics :192 quarterly observations, no missing values					
Series	N	Min	Median	Mean	Max
Real RPPI (2010=100)	192	35.9	63.5	74.8	141.8
10-yr Bond Yield (%)	192	0.88	5.83	7.21	16.03
Unemployment (%)	192	3.5	6.2	6.5	11.1



Do the Macro Drivers Explain RPPI?

RPPI vs Bond Yield

RPPI vs Unemployment



When yields fell, property prices climbed. The negative relationship holds across the full 47-year sample.



Stationarity Testing & Box-Cox Transformation

Three formal tests confirm $d=1$ · $\lambda \approx 0$ log-like transformation applied

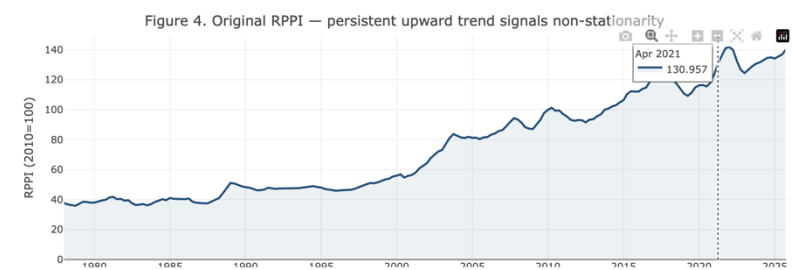
Series	Test	Null Hypothesis	Statistic	p-value	Decision
Levels	ADF	Unit root (non-stationary)	-2.317	0.444	Non-stationary
Levels	PP	Unit root (non-stationary)	-10.08	0.540	Non-stationary
Levels	KPSS	Stationary	0.677	<0.010	Non-stationary
1st Diff (BC)	ADF	Unit root (non-stationary)	-6.402	<0.010	Stationary
1st Diff (BC)	PP	Unit root (non-stationary)	-89.86	<0.010	Stationary
1st Diff (BC)	KPSS	Stationary	0.057	>0.100	Stationary

Box-Cox $\lambda = -0.0102$

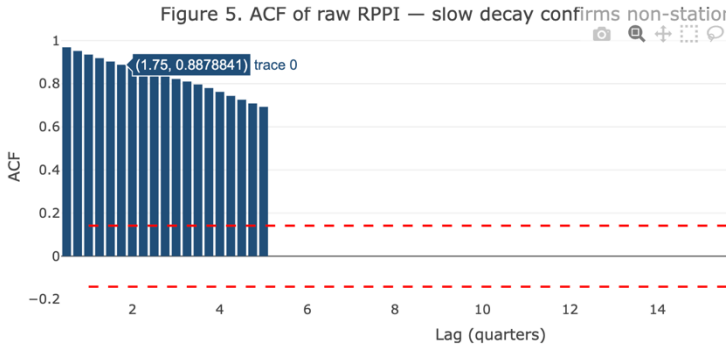
$\lambda \approx 0 \rightarrow$ log-like transformation
Applied to stabilise variance
before differencing

Decision: $d = 1$

All 3 tests agree after first
difference. KPSS $p > 0.10$
 $\rightarrow$ stationarity confirmed



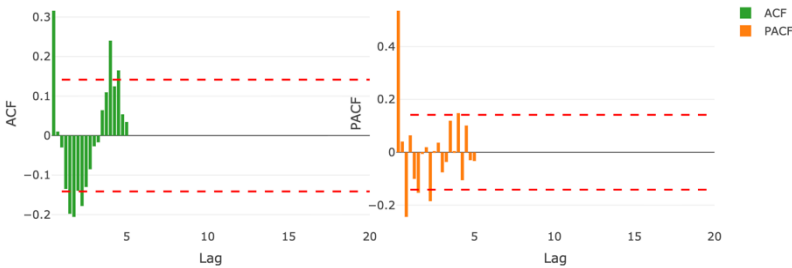
Conclusion: Persistent upward trend + slowly decaying ACF = non-stationary. Formal tests required.



No Over-differencing

Differenced series retains
usable ARMA structure in
ACF/PACF (checked next slide)

After Differencing



Figures 10-11. ACF and PACF of differenced series structure present

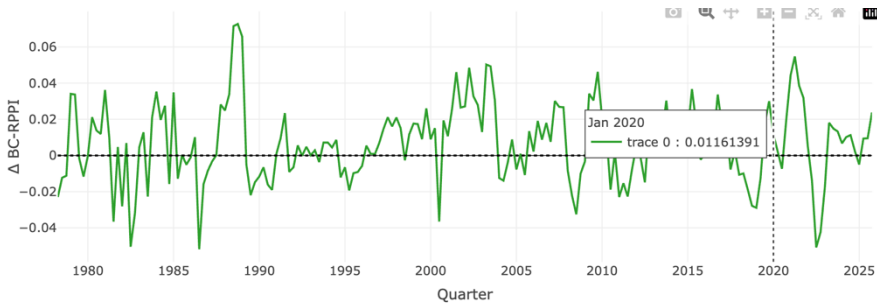


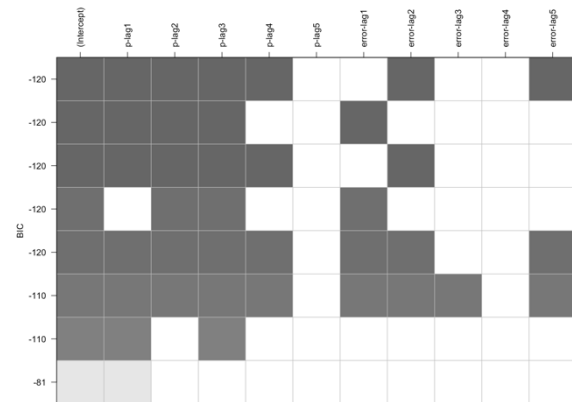
Figure 9. First-differenced Box-Cox RPPI stationary, fluctuates around zero



Model Comparison (AIC & BIC)

EACF Output : eacf(rppi_diff, ar.max=6, ma.max=6)

AR/MA	0	1	2	3	4	5	6
0	x	x	o	o	o	x	x
1	o	x	o	o	o	o	o
2	x	x	o	o	o	o	o
3	x	x	x	x	o	o	o
4	x	x	x	x	o	o	o
5	x	x	x	x	o	o	o
6	o	o	x	x	o	o	o



- Top-left cluster of 'o' at (AR=0, MA=2) and (AR=1, MA=2) → MA(2) structure dominant. Candidates: ARIMAX(0,1,1), (1,1,1), (2,1,1), (1,1,2), (2,1,2).
- BIC table (armasubsets) confirms: ARIMAX(1,1,2) = p-lag1 + error-lag1 + error-lag2 appears at BIC ≈ -121, 4th row from top

Overfitting Check :

- **ar1 and ar2 both insignificant** ($p > 0.49$) in the overfit model. BIC worsens from -973.8 to -969.1. Parsimony confirmed
- ARIMAX(1,1,2) is the correct choice

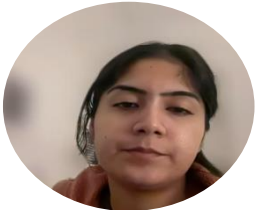
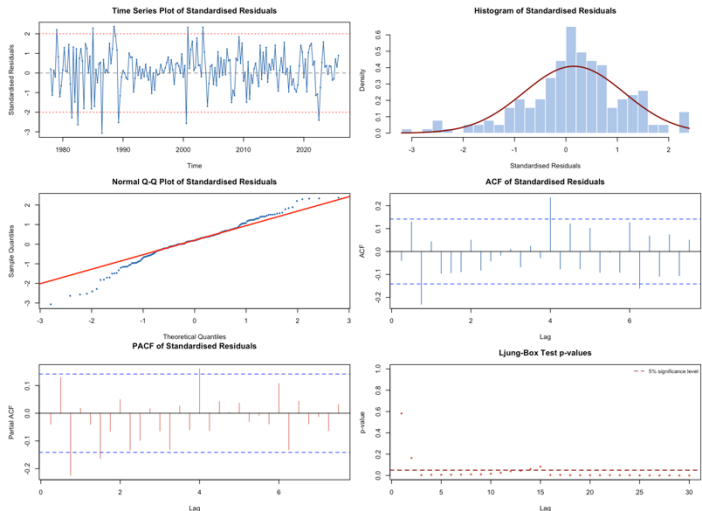
Tables 3 & 4. AIC & BIC : Five Candidate ARIMAX Models

Model	df	AIC	BIC	Rank (BIC)
ARIMAX(1,1,2)	6	-993.325	-973.812	1st
ARIMAX(2,1,1)	6	-989.897	-970.383	2nd
ARIMAX(2,1,2)	7	-991.846	-969.080	3rd
ARIMAX(1,1,1)	5	-981.318	-965.056	4th
ARIMAX(0,1,1)	4	-956.473	-943.464	5th

Table 5. ARIMAX(1,1,2) : coeftest() Output (ML)

Parameter	Estimate	Std. Error	z-value	p-value	Sig.
ar1	-0.0856	0.1449	-0.590	0.5549	-
ma1	0.6493	0.1240	5.235	<0.001	***
ma2	0.5147	0.0837	6.152	<0.001	***
bond10	0.0017	0.0020	0.850	0.3954	(kept)
unemp	-0.0116	0.0044	-2.610	0.0091	**

ar1 insignificant but MA(2) structure and unemployment are strongly significant.

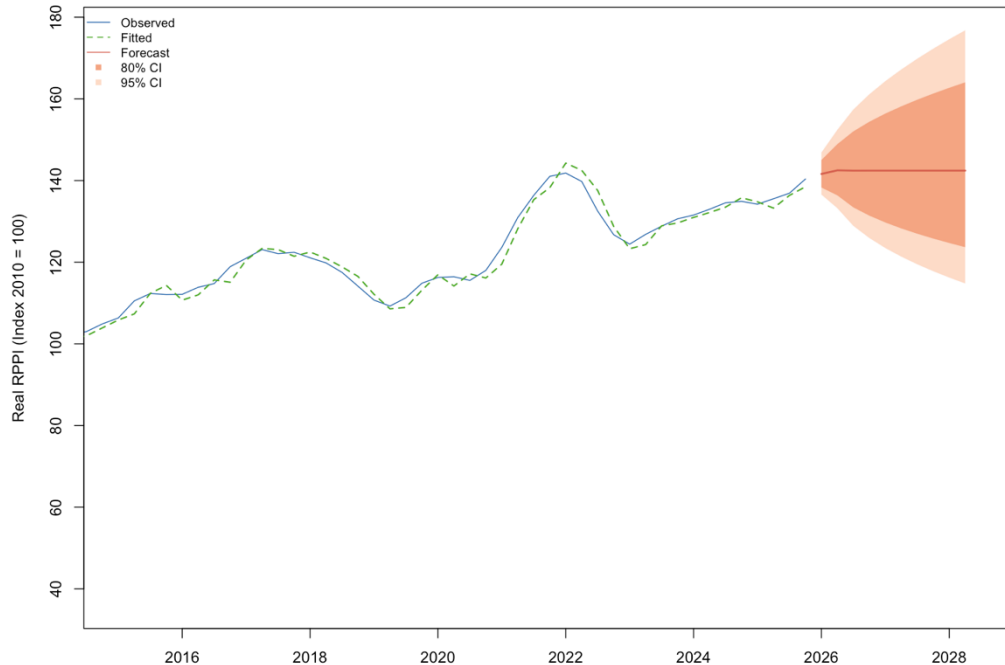


10-Quarter Base Forecast

Model: ARIMAX(1,1,2) · MAPE = 1.37% · RMSE = 1.33 · Bond yield held at 4.46% · Unemployment at 4.25%



Figure 13. 10-Quarter Forecast – ARIMAX(1,1,2)
(Q1 2026 to Q2 2028)



Model Accuracy (In-Sample)

MAPE: 1.37%

Mean Absolute Percentage Error

RMSE: 1.332

Root Mean Squared Error

MAE: 0.988

Mean Absolute Error

ME: 0.254

Mean Error (bias)

Interpretation of the Forecast

- Base forecast stabilises near 142.4 from Q2 2026 onwards slight plateau after strong post-COVID run
- Prediction intervals widen from ± 3.4 (Q1 2026) to ± 20.2 (Q2 2028) uncertainty compounds over time
- Q1 2026 forecast of 141.6 is closest to recent observed values near-term forecasts most reliable
- Widening 95% CI reflects macro uncertainty, not model inadequacy residuals are white noise
- MAPE of 1.37% indicates excellent in-sample fit across 192 observations
- Bond yield held constant at 4.46% any RBA cut would shift the base path upward

Table 9. 10-quarter point forecasts with 80% and 95% prediction intervals

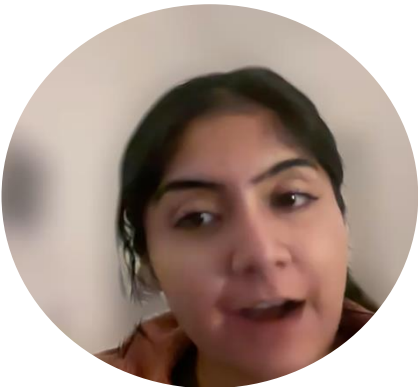
Quarter	Forecast	Lo_80	Hi_80	Lo_95	Hi_95
2026 Q1	141.62	138.29	145.02	136.57	146.85
2026 Q2	142.5	136.36	148.93	133.22	152.44
2026 Q3	142.43	133.43	152.04	128.9	157.4
2026 Q4	142.43	131.41	154.4	125.92	161.13
2027 Q1	142.43	129.73	156.4	123.47	164.34
2027 Q2	142.43	128.27	158.18	121.36	167.21
2027 Q3	142.43	126.97	159.8	119.49	169.84
2027 Q4	142.43	125.79	161.3	117.79	172.29
2028 Q1	142.43	124.7	162.71	116.24	174.61
2028 Q2	142.43	123.69	164.05	114.8	176.81

Conclusion

- **Bond yields and unemployment significantly explain property prices:** unemployment significant at $p < 0.05$; bond yield retained as structural regressor
- **ARIMAX(1,1,2) selected :** lowest AIC (-993.325) and BIC (-973.812) across five candidates
- **Residuals pass all tests :** Shapiro-Wilk $p = 0.084$ (normal); Ljung-Box $p > 0.05$ (no remaining autocorrelation)
- **Base forecast projects prices near 142.4** over 10 quarters : with widening uncertainty bands as the horizon extends
- **Scenario analysis shows +0.6 (bull) / -1.1 (bear) spread by Q2 2028** directly driven by RBA policy and labour market assumptions

Take Away

The ARIMAX model is the first layer of a property investment decision.
It tells you which direction macro conditions are pushing prices.
What it cannot tell you is whether a specific property, in a specific suburb, at a specific asking price is a good buy.
But if the model says the macro tailwind is turning into a headwind that is always worth knowing first



Three Scenarios: Time Series the way it can be used in future

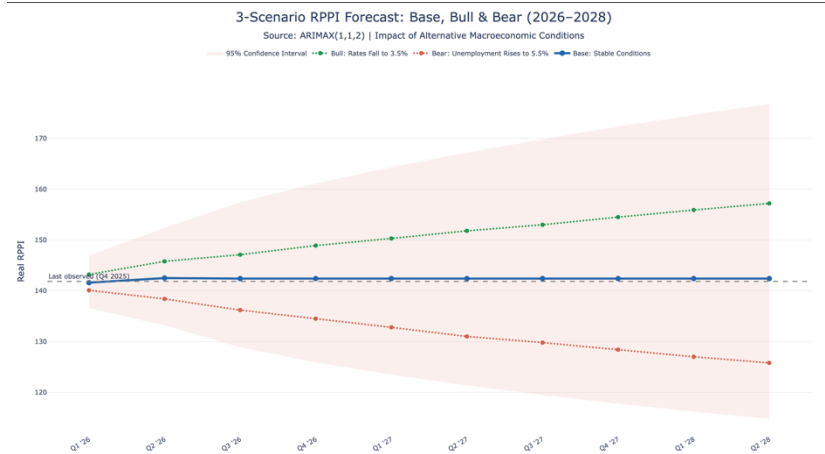


Table 10. Scenario forecast outcomes and investor signals			
Scenario	Macro Assumption	Q2_2028_Forecast	Investor Signal
Bull	RBA cuts -1pp, unemployment falls to 3.8%	143.1	Strong buy enter now
Base	Conditions stable at Q4 2025 levels	142.4	Wait and watch
Bear	RBA hikes +1pp, unemployment rises to 5.1%	141.3	Defensive reduce exposure